

Abstract

Modern cyber-attacks such as DDoS and zero-day exploits evolve faster than human operators can respond, making Mean Time to Respond (MTTR) the critical metric for network defense. This project proposes a decentralized multi-agent system (MAS) for automated, collaborative incident response. The architecture distributes defensive responsibilities across five specialized agent types: Traffic Monitoring, Anomaly Classification, Response Coordination, Threat Intelligence, and Resource Allocation. Operating within a segmented network, these agents utilize an asynchronous publish-subscribe bus to coordinate actions in real-time. Advanced interaction models, including coalition formation for multi-segment threats, auction-based resource distribution, and majority voting for high-risk escalations, ensure proportional and efficient defense. The system targets a detection rate above 90%, service availability over 99%, and an MTTR under 1,000 milliseconds. Validated through six stress-test scenarios covering agent failure and novel attack emulation, this MAS framework offers a scalable, resilient solution for security in dynamic environments

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1. Introduction

1.1. Overview

Modern cyber-attacks are increasingly fast, coordinated, and adaptive, rendering traditional security tools insufficient for timely detection and mitigation. Manual incident-response processes frequently fail because attacks such as DDoS floods, ransomware campaigns, and zero-day exploits evolve faster than human operators can perceive or neutralize them.

Consequently, Mean Time to Respond (MTTR) becomes the decisive metric, any delay in containment can cause severe integrity loss, service disruption, and financial damage.

This project proposes a multi-agent system that enables collaborative cyber-defense, wherein autonomous agents jointly monitor, detect, analyze, and respond to threats within a decentralized network. The design follows established MAS engineering principles to guarantee scalability, robustness, and provable convergence of defensive actions.

1.2. Goals

The primary goal is to enhance cyber-defense effectiveness by distributing responsibilities across specialized agents rather than relying on a monolithic controller. Each agent will assume a focused role in traffic monitoring, anomaly classifier, threat-intelligence sharing, response coordination, or resource allocation, and will cooperate through coalition formation, auction-based resource distribution, and voting-based escalation to synthesize a coherent defense strategy.

1.3. Scope

The scope encompasses the full lifecycle of a multi-agent cyber-defense system: requirements analysis, architectural design, agent modeling, communication protocol specification, and validation through simulated attack scenarios.

1.4. Glossary

- DDoS - Distributed Denial-of-Service attack, overwhelming a target with traffic.
- Ransomware - Malware that encrypts data and demands payment for decryption.
- Zero-day exploit - An attack leveraging a previously unknown vulnerability.
- Mean Time to Respond (MTTR) - Average elapsed time from attack detection to successful mitigation.
- Lateral Movement - A technique by which an attacker traverses a network from an initially compromised host to other machines, typically to escalate privileges or reach higher-value targets.
- Port Scanning - Systematic probing of a host or range of hosts to identify open network ports and the services listening on them.
- Quarantine - A defensive action that isolates a compromised host or entire network segment to prevent the spread of an attack.
- Topic (Pub/Sub) - A named channel on the message bus to which agents can publish messages or subscribe to receive them (alerts, threat-reports, resource-bids).

2. Environment Specification

Cyber defense is not a static problem. Attackers adapt, traffic fluctuates, and no single vantage point reveals the full picture of what is happening across a network. To study how autonomous agents might respond to this reality, the simulation environment must be carefully specified - not only in terms of what it contains, but in terms of how it behaves. The environment specification is therefore organized into two parts: a classification that captures its behavioral and structural properties, and a boundary definition that delimits its physical and logical scope.

Classification

The environment can be classified as follows:

- Mixed Collaborative/Competitive: Defense agents cooperate with each other to protect the network, while attackers act as opposing entities trying to disrupt services, evade detection, or compromise resources.
- Partially Observable: Each agent observes only a local subset of the system, such as a network segment, endpoint status, alert stream, or traffic pattern.
- Dynamic/Non-Stationary: The environment evolves continuously as traffic loads change, users generate new activity, and adversaries adapt their tactics.
- Stochastic: The behavior of the system is probabilistic, meaning identical actions may lead to different outcomes depending on traffic conditions, attack intensity, and timing.
- Continuous: Attacks can happen at any moment; network state changes in real time. Agents must be ready to act at any point, not just at fixed time steps.

Boundaries

The environment represents a simulated corporate or institutional network divided into four distinct segments. Each segment is a zone of the network with its own security profile and threat surface:

- Internal User Subnet: Represents the area of the network used by regular employees. Threats here often involve ransomware, phishing, or lateral movement - where an attacker moves from one machine to another.
- Server Zone: Houses critical servers such as databases, application servers, and internal tools. A high-value target for attackers seeking sensitive data.
- Public-Facing Services: The part of the network exposed to the internet, such as web servers and APIs. Most susceptible to DDoS attacks and exploitation of public-facing vulnerabilities.
- Security Monitoring Zone: A protected area where monitoring tools, IDS alerts, and threat intelligence feeds operate.

Each segment represents a distinct observation and action region for one or more agents, allowing the system to model localized threats and coordinated response actions.

The environment excludes physical hardware implementation details and focuses instead on the logical cyber-defense layer. The scope is limited to the simulated network, the data exchanged between agents, and the response mechanisms executed within the host environment. The host environment constraints include available CPU, memory, and simulation time, which limit how many agents can operate simultaneously and how complex their learning, detection, and coordination mechanisms can be. Because of these constraints, the system must remain computationally efficient while still supporting real-time monitoring, agent communication, and visualization of network state.

3. Agent Specification

This section defines each type of agent in the system: what its purpose is, what it can perceive (its inputs), and what actions it can take (its outputs).

3.1. Defense Agents

There are five types of defense agents. Multiple instances of each type may run simultaneously for instance, one Traffic Monitor agent per network segment.

3.1.1. Agent Type 1: Traffic Monitor Agent (TMA)

Role:

Acts as the eyes of the network. Continuously reads raw network traffic in its assigned segment and looks for patterns that deviate from normal behavior.

Perception:

- Raw packet-level data: source IP, destination IP, port numbers, protocol, packet size, and frequency.
- Current traffic volume compared to a historical baseline (the 'normal' level).
- Timestamps of all observed events.

Actions:

- Publish an alert message to the shared communication bus when anomalous traffic is detected.
- Update its local traffic baseline model using the most recent data.
- Request additional monitoring resources from the Resource Allocator if traffic load is very high.

Goal:

Maximize the detection rate of genuine anomalies in its segment while keeping false positives to a minimum.

3.1.2. Agent Type 2: Anomaly Classifier Agent (ACA)

Role:

The analyst. Takes raw alerts from Traffic Monitor Agents and applies machine learning classification to determine whether an anomaly is truly a threat, and if so, what kind.

Perception:

- Alert messages published by Traffic Monitor Agents.
- Threat intelligence data shared by other Anomaly Classifier Agents.
- Its own local classification model (for instance clustering of known attack signatures).

Actions:

- Classify an alert as: Normal, Suspicious, or Confirmed Threat.
- Assign a threat severity score (0.0 = harmless, 1.0 = critical).
- Broadcast a threat report to the communication bus with its classification result.
- Update its local model using feedback from past incidents (online learning).

Goal:

Accurately classify threats and continuously improve classification accuracy over time.

3.1.3. Agent Type 3: Response Coordinator Agent (RCA)

Role:

The incident commander. When a confirmed threat is reported, the RCA determines the best defensive response and coordinates other agents to execute it.

Perception:

- Threat reports from Anomaly Classifier Agents.
- Current defensive resource availability (from the Resource Allocator).
- Voting results from a coalition (when a high-risk action requires group approval).
- Available response action options and their estimated effectiveness.

Actions:

- Issue a response directive: traffic throttling, port blocking, agent redeployment, or segment quarantine.
- Initiate a voting round among coalition members before executing high-risk actions.
- Request resource allocation via the auction mechanism.
- Log all decisions and their outcomes for post-incident analysis.

Goal:

Minimize MTTR (Mean Time To Respond) while maximizing system availability, using the most proportional and effective response.

3.1.4. Agent Type 4: Threat Intelligence Agent (TIA)

Role:

The memory and knowledge broker of the coalition. While all other agents operate within their local segment, the TIA looks across all segments simultaneously, correlating what different agents are seeing to determine whether isolated observations are part of a larger, coordinated attack. It serves as the institutional memory of the system, retaining historical incident data and ensuring that what one agent learns during an incident is not lost when it resolves, but is folded back into the shared knowledge base every agent draws

Perception:

- Threat reports from all Anomaly Classifier Agents across all segments.
- External threat feeds (simulated known attack signatures).
- Historical incident logs.

Actions:

- Publish enriched threat intelligence summaries to the shared bus.
- Identify attack patterns that span multiple segments (e.g., coordinated lateral movement).
- Trigger coalition formation when a multi-segment threat is detected.
- Score and rank active threats by urgency for the Resource Allocator.

Goal:

Ensure no agent operates in isolation with outdated information. Maximize the quality and timeliness of shared intelligence by maintaining a continuously updated, system-wide threat picture that individual agents could not construct on their own.

3.1.5. Agent Type 5: Resource Allocator Agent (RAA)

Role:

The logistics manager. Manages shared defensive resources, such as deep packet inspection capacity, quarantine slots, and computational budget, and distributes them efficiently using an auction mechanism.

Perception:

- Incoming resource bids from all agents.
- Current resource availability and utilization levels.
- Threat severity scores (used to prioritize allocation).

Actions:

- Run a sealed-bid auction: collect bids, determine winners based on threat priority, and allocate resources accordingly.
- Notify winning and losing agents of their allocation outcome.
- Revoke previously allocated resources from resolved incidents and redistribute them.

Goal:

Maximize system-wide resource efficiency, ensure the highest-severity threats always get the resources they need first.

3.2. Attacker Agents (Adversaries)

Attacker agents exist to stress-test the defense system. They are not real attackers but simulated programs that follow realistic attack strategies.

Attacker Type	Behavior Description
DDoS Attacker	Floods one or more network segments with high volumes of traffic. Aims to exhaust bandwidth and make services unavailable. Gradually increases intensity to probe defenses.
Port Scanner	Systematically probes the network for open ports and vulnerable services. Has low traffic volume but very structured patterns.
Lateral Movement Agent	Simulates an attacker who has already breached the internal network and is moving quietly from machine to machine, looking for higher-privilege targets.
Zero-Day Emulator	Generates traffic patterns that do not match any known attack signature. Used to test whether Anomaly Classifier agents can identify novel threats without prior examples.

4. Functional Requirements

The functional requirements section outlines the core capabilities of the cyber defence system.

4.1. Individual Agent Requirements

4.1.1. Traffic Monitor Agent (TMA)

ID	Requirement
FR-01	The TMA shall continuously sample network traffic in its assigned segment at a minimum rate of 10 times per second.
FR-02	The TMA shall compute a rolling average traffic baseline and flag any observation that deviates by more than 2 standard deviations from the baseline as an anomaly.
FR-03	The TMA shall publish an alert message within 100 milliseconds of detecting an anomaly.
FR-04	The TMA shall update its baseline model at least once every 60 seconds using recent traffic data.

4.1.2. Anomaly Classifier Agent (ACA)

ID	Requirement
FR-05	The ACA shall classify every received alert as Normal, Suspicious, or Confirmed Threat within 200 milliseconds of receipt.
FR-06	The ACA shall assign a numeric severity score between 0.0 and 1.0 to every Confirmed Threat, where 1.0 represents the highest urgency.
FR-07	The ACA shall broadcast a structured threat report to the shared communication bus for every Confirmed Threat or Suspicious event.
FR-08	The ACA shall update its classification model after every resolved incident to incorporate new patterns (online learning).
FR-09	The ACA shall maintain a local False Positive Rate below 10% across any rolling 10-minute window of traffic.

4.1.3. Response Coordinator Agent (RCA)

ID	Requirement
FR-10	The RCA shall initiate a defensive response within 500 milliseconds of receiving a Confirmed Threat report with <i>severity</i> ≥ 0.7 .
FR-11	The RCA shall initiate a coalition voting round before executing any quarantine action. The action shall only proceed if a simple majority (>50%) of coalition members vote to approve.
FR-12	The RCA shall log every decision (threat received, response chosen, outcome) to a persistent incident log.
FR-13	The RCA shall select the least disruptive response action that is expected to neutralize a threat (proportionality principle).
FR-14	The RCA shall send a resolution notification to all coalition members when a threat is neutralized.

4.1.4. Threat Intelligence Agent (TIA)

ID	Requirement
FR-15	The TIA shall aggregate threat reports from the ACA agent and construct a global threat model updated at least every 500 milliseconds.
FR-16	The TIA shall detect when two or more segments are reporting correlated threats and flag this as a multi-segment incident within 1 second.
FR-17	The TIA shall trigger coalition formation automatically when a multi-segment incident is detected.
FR-18	The TIA shall rank all active threats by severity and publish an updated priority list to all agents every 1 second.

4.1.5. Resource Allocator Agent (RAA)

ID	Requirement
FR-19	The RAA shall conduct a sealed-bid auction within 300 milliseconds of receiving resource bids from two or more competing agents.
FR-20	The RAA shall prioritize resource allocation to the agent with the highest-severity active threat, using the threat score as the primary bid evaluation criterion.
FR-21	The RAA shall notify all bidding agents of the auction outcome within 100 milliseconds of the auction conclusion.
FR-22	The RAA shall reclaim and redistribute resources from resolved incidents within 500 milliseconds of receiving a resolution notification.
FR-23	Total system resource usage managed by the RAA shall never exceed 40% of the host machine's CPU and memory capacity.

4.1.6. Attackers

ID	Requirement
FR-24	The DDoS Attacker shall distribute its traffic across randomized source IPs to simulate a botnet and reduce the effectiveness of simple IP-based blocking.
FR-25	The Port Scanner shall vary its scan order pseudo-randomly across runs to avoid producing a fixed, easily-identified pattern.
FR-26	The Zero-Day Emulator shall generate traffic that does not match any signature in the ACA's classification model, verified by confirming no existing rule or cluster centroid covers the generated pattern.
FR-27	All attacker agents shall log every action taken, target segment, traffic volume, timing, and whether a defensive response was triggered, to enable post-simulation analysis.
FR-28	Attacker agents shall operate independently of one another unless explicitly configured for a coordinated attack scenario, in which case a shared attack schedule shall synchronize their actions.

4.2. System-Level Requirements

ID	Requirement
FR-29	The overall system shall achieve a Detection Rate of at least 90% across all attack types in the validation scenarios.
FR-30	The system shall maintain a Mean Time to Respond (MTTR) below 1,000 milliseconds for all Confirmed Threats with <i>severity</i> ≥ 0.7 .
FR-31	The system shall maintain protected service availability above 99% during all simulated attack scenarios.
FR-32	All inter-agent messages shall follow a structured schema. Agents shall reject malformed messages and log the rejection.
FR-33	The system shall support a minimum of 5 simultaneous active incidents without performance degradation.
FR-34	The system shall be resilient to the loss of a single agent - if one agent goes offline, the remaining agents shall continue to function and take over its duties within 2 seconds.

5. Simulation and Visualization Requirements

The simulation environment serves as the primary verification tool for the multi-agent system. It provides a controlled, real-time environment where an observer can validate that agents are adhering to their defined defensive strategies and interaction protocols.

5.1. Simulation Platform Requirements

The simulation platform provides the underlying logic and data generation necessary to stress-test the defensive agents.

- Segmented Topology Modeling: The simulator shall model the four network segments defined in the environment specification: the Internal User Subnet, Server Zone, Public-Facing Services, and the Security Monitoring Zone.
- Synthetic Traffic Generation: The platform shall generate synthetic network traffic for each segment at configurable volumes, simulating both standard user behavior and malicious traffic patterns.
- Dynamic Attack Injection: The simulator shall support the manual or automated injection of specific attack types, including DDoS, Port Scans and Zero-Day Emulations.
- Configurable Temporal Scaling: To facilitate observation, the simulator shall run in real time or at an accelerated/decelerated speed multiplier.
- Programmatic Interface (API): The platform shall expose a programmatic API allowing agents to read raw packet data and execute response actions like port blocking or quarantining.
- Comprehensive Event Logging: Every agent decision, inter-agent message, and environment state change shall be recorded in a persistent event log for post-run analysis.

5.2. Visualization Requirements

The visualizer acts as the human-machine interface, translating abstract agent communications and network states into a legible format.

- Interactive Network Topology Map: The system shall display the four segments as nodes on a graphical map. Individual agents must be represented by icons within their assigned segments, using color-coding to reflect segment health: Green (Normal), Orange (Suspicious), and Red (Confirmed Threat).
- Real-Time Agent Status Panel: An inspector panel shall display the live state of any selected agent, including its current task, the local threat score it is processing, and its most recent action taken.
- Chronological Alert Feed: A live scrolling feed shall display all published alerts and threat reports, annotated with the originating agent's ID and the severity level assigned by the Anomaly Classifier Agent.
- Message Flow Visualization: Communication between agents (bids, votes, or intelligence sharing) shall be visualized as arrows between agent icons on the network map.
- Coalition & Grouping Overlay: When a coalition is formed to handle a multi-segment incident, the participating agents shall be visually highlighted or connected to indicate coordinated action.
- Resource Allocation Dashboard: The visualizer shall provide a real-time bar chart showing the computational and defensive resources managed by the Resource Allocator Agent, including active auction outcomes and remaining capacity.
- Live Performance Metrics: A panel that graphs the system's primary success metrics in real time: Detection Rate, False Positive Rate, System Availability, and Mean Time to Respond (MTTR).

5.3. Observer Verification Support

To allow an observer to confirm that agents are adhering to their defined strategies, the platform shall provide the following verification aids:

- Strategy Trace: Each agent shall emit a structured reasoning log entry for every decision, containing the inputs received, the rule or utility calculation applied, and the action chosen. This log shall be human-readable and exportable.
- Constraint Checker: An automated post-run validator shall compare the strategy trace against the functional requirements and flag any violation. For example, an ACA that exceeded the 200 ms classification deadline or an RCA that skipped the voting step before a quarantine action.
- Replay Mode: The simulation shall support replaying any recorded run at any speed, with the visualization synchronized to the event log, so the observer can step through the timeline of a specific incident.

6. Interaction Models

Interaction models define the structured protocols through which agents communicate, cooperate, and resolve conflicts. These mechanisms ensure that the decentralized system achieves social welfare (defined in this system as maximum network security and availability) without requiring a single point of failure.

6.1. Publish-Subscribe (Pub/Sub) Communication Protocol

The system utilizes a decoupled asynchronous communication model via a central message bus. This allows agents to operate without direct knowledge of one another, enhancing system scalability and fault tolerance.

- Mechanism: A central message bus acts as the communication backbone.
- Subscriptions: Agents subscribe to specific "topics" based on their roles. For example, the Anomaly Classifier Agent (ACA) subscribes to the alerts topic, while the Response Coordinator Agent (RCA) subscribes to threat-reports.
- Publication: When a Traffic Monitor Agent (TMA) detects an anomaly, it publishes a message to the alerts topic for immediate receipt by all relevant ACAs.

6.2. Coalition Formation

When threats span multiple network segments or exceed the capabilities of a single agent, the system initiates Coalition Formation to create a temporary, goal-oriented team.

- Trigger: The Threat Intelligence Agent (TIA) detects correlated threats across two or more segments.
- Announcement: The TIA broadcasts a coalition invitation specifying the threat ID and required expertise.
- Response: Available agents (those not currently engaged in high-priority tasks) must accept or decline within 200ms.
- Confirmation: The TIA confirms membership and designates a lead RCA.
- Dissolution: Upon threat neutralization, the TIA broadcasts a dissolution message, and agents return to independent monitoring.
- Constraint: To prevent resource over-utilization, an agent may participate in at most one coalition at any given time.

6.3. Auction-Based Resource Allocation

To manage limited defensive resources - such as deep packet inspection (DPI) capacity or quarantine slots, the system employs a sealed-bid, first-price auction managed by the Resource Allocator Agent (RAA).

- Bidding: Agents submit private bids where the bid value B is a function of the threat severity score S , such that $B = S$ (where $0.0 \leq S \leq 1.0$).
- Evaluation: The RAA ranks bids and allocates resources to the highest bidders until capacity is reached.
- Resolution: Winning agents receive resources for a fixed duration or until the incident is resolved, at which point resources are reclaimed for redistribution.

6.4. Voting-Based Escalation Protocol

To prevent service disruptions caused by false positives, high-impact defensive actions, such as segment-wide quarantine, require collective authorization through a majority consensus.

- Proposal: The RCA initiates a vote within the active coalition before executing a quarantine action.
- Evaluation: Coalition members evaluate the proposal against their local observations (current traffic health).
- Threshold: A majority vote (where $\text{Votes} > 50\%$) is required to proceed.
- Temporal Constraint: The entire voting cycle must conclude within 300ms to maintain the system's MTTR targets.

6.5. Attacker-Defender Game Model

The interaction between simulated adversaries and the defense coalition is modeled as a two-player zero-sum game. This framework acknowledges the competitive nature of the environment where an attacker's gain in network disruption is directly proportional to the defender's loss in system availability.

- Best-Response Strategy: Defense agents utilize observed attack patterns to compute the most effective countermeasures (if a DDoS is detected, the "best response" is to increase throttling resources via auction).
- Mixed Strategies: To prevent attackers from predicting and bypassing defenses, the system can rotate defensive actions, creating uncertainty for the adversary.
- Feedback Loop: Results of these interactions are logged by the TIA and used to update the ACA's classification models, effectively "learning" the attacker's playbook over time.

7. Utility and Success Metrics

This section provides the formal mathematical framework for evaluating agent performance and overall system effectiveness. In this multi-agent system, utility functions serve as the primary mechanism for quantifying how well autonomous entities achieve their local goals while contributing to the global security objective.

7.1. Individual Agent Utility Functions

Each agent type operates according to a specific utility function designed to maximize its performance, either by contributing to the collective defense of the network or by achieving specific adversarial objectives during an attack.

7.1.1. Traffic Monitor Agent (TMA) Utility

The TMA utility focuses on the accuracy and speed of the initial Anomaly Classifier within its assigned segment.

$$U_{TMA} = DR_{segment} \cdot (1 - FPR_{segment}) \cdot \left(\frac{1}{MTTR_{alert}} \right)$$

- $DR_{segment}$: Detection Rate - the fraction of total simulated attacks correctly flagged by the TMA.
- $FPR_{segment}$: False Positive Rate - the fraction of Normal traffic incorrectly identified as an anomaly.
- $MTTR_{alert}$: Mean Time to Respond (Alert) - the time elapsed between anomaly occurrence and alert publication, with a target of <100 ms.

7.1.2. Anomaly Classifier Agent (ACA) Utility

The ACA utility measures the precision of threat classification and the effectiveness of its online learning mechanisms.

$$U_{ACA} = accuracy_{classification} \cdot (1 - FPR_{ACA}) \cdot model_improvement_rate$$

- $accuracy_{classification}$: The ratio of correctly classified threats (True Positives + True Negatives) to total alerts received.
- FPR_{ACA} : The false positive rate specifically at the classification stage, required to remain below 10%.
- $model_improvement_rate$: A metric tracking the delta in classification accuracy across successive simulation episodes, reflecting the agent's "online learning" goal.

7.1.3. Response Coordinator Agent (RCA) Utility

The RCA utility prioritizes maintaining service availability while ensuring defensive actions are swift and proportional.

$$U_{RCA} = availability \cdot \left(\frac{1}{MTTR_{Response}} \right) \cdot proportionality_score$$

- *availability*: The fraction of time that critical services remain reachable by legitimate users.
- $MTTR_{Response}$: Mean Time to Respond (Action) - the time from receiving a confirmed threat to executing a mitigation.
- *proportionality_score*: A qualitative score (0.0 to 1.0) assessing if the agent selected the least disruptive response necessary to neutralize the threat.

7.1.4. Resource Allocator Agent (RAA) Utility

The RAA utility evaluates the efficiency of the auction mechanism and the system's impact on the host environment.

$$U_{RAA} = resource_efficiency \cdot (1 - resource_overhead)$$

- *resource_efficiency*: The ratio of resources successfully allocated to high-severity threats ($S \geq 0.7$) versus total allocations.
- *resource_overhead*: The total CPU and memory consumption of the MAS relative to the host machine's capacity, which must not exceed 40%.

7.1.5. Threat Intelligence Agent (TIA) Utility

The TIA utility measures the quality and timeliness of shared intelligence, and the agent's effectiveness in coordinating multi-segment threat response.

$$U_{TIA} = intelligence_coverage \cdot correlation_accuracy \cdot \left(\frac{1}{MTTR_{Coalition}} \right)$$

- *intelligence_coverage*: The fraction of active threats across all segments for which the TIA has published an enriched threat summary within the required 500 ms update window. A value of 1.0 means every active threat was covered on time.
- *correlation_accuracy*: The ratio of correctly identified multi-segment incidents (true coalitions formed) to all coalition triggers issued. Penalizes false coalitions formed on uncorrelated threats.
- $MTTR_{Coalition}$: The mean time from multi-segment threat detection to confirmed coalition formation, with a target of < 1,000 ms.

7.1.6. Attacker Agent Utility

The Attacker Agent utility quantifies the effectiveness of an adversary in disrupting services and evading detection. In the mixed collaborative/competitive environment, this utility serves as the inverse performance metric for the defense coalition.

$$U_{ATK} = disruption_impact \cdot evasion_rate \cdot \left(1 - \frac{1}{MTTR_{Response}} \right)$$

- *disruption_impact*: The magnitude of service degradation achieved, measured by the reduction in protected service availability. A value of 1.0 represents a total denial-of-service (0% availability) for the duration of the attack.
- *evasion_rate*: The fraction of attack duration during which the attacker remains unmitigated. This is calculated as the time elapsed before a "Confirmed Threat" status is broadcast by the Anomaly Classifier Agent (ADA).
- $MTTR_{Response}$: The system's Mean Time to Respond. From the attacker's perspective, a higher MTTR indicates a more successful exploitation window, whereas a low MTTR (approaching the 1,000 ms target) minimizes the attacker's utility.

7.2. Social Welfare Function

Social Welfare (SW) represents the aggregate performance of the multi-agent defence system. It is calculated as a weighted sum of individual utilities to reflect the relative importance of each role in the defense lifecycle.

$$SW = \sum_{i=1}^n w_i \cdot U_i (w_1 \cdot U_{TMA})(w_2 \cdot U_{ACA})(w_3 \cdot U_{RCA})(w_4 \cdot U_{RAA})(w_5 \cdot U_{TIA})$$

Agent	Weight	Justification
TMA (Traffic Monitor)	$w_1 = 0.2$	Critical for initial visibility, though accuracy is verified by the ACA.
ACA (Anomaly Classifier)	$w_2 = 0.3$	The most critical function, classification errors lead to improper response or service loss.
RCA (Response Coordinator)	$w_3 = 0.25$	Directly responsible for MTTR and availability targets.
RAA (Resource Allocator)	$w_4 = 0.1$	Supporting function ensuring system stability and resource fairness.
TIA (Threat Intelligence)	$w_5 = 0.15$	Facilitates coalition formation and handles multi-segment threats.

Success Threshold: The system is considered validated and successful if the overall Social Welfare score is $SW \geq 0.80$ across all validation scenarios.

7.3. Summary of Quantitative Targets

The following table summarizes the mandatory performance requirements that the system must satisfy as demonstrated through the simulation.

Metric	Definition	Formula / Measure	Target	Agent(s)
Detection Rate (DR)	Fraction of real attacks correctly identified	True Positives / All Attacks	> 90%	TMA, ACA
False Positive Rate (FPR)	Fraction of benign traffic falsely flagged	False Positives / All Normal	< 8%	TMA, ACA
MTTR (Alert)	Time from anomaly to alert published	Milliseconds	< 100 ms	TMA
MTTR (Response)	Time from confirmed threat to action	Milliseconds	< 1000 ms	RCA
System Availability	Uptime of protected services during attack	Uptime / Total Time	> 99%	RCA, RAA
Resource Overhead	MAS CPU+memory usage on host	% of host capacity	< 40%	RAA
Coalition Efficiency	Ratio of threats needing multi-agent response	Multi-agent / Total Incidents	Monitored	TIA, RCA
Social Welfare (SW)	Overall system performance score	Weighted utility average	>= 0.80	All agents

8. Validation Scenarios

Each scenario is executed within the simulation environment to evaluate the system against the quantitative targets established in the success metrics. The addition of Attacker Utility (U_{ATK}) allows for the measurement of adversarial success in direct competition with the defense coalition's Social Welfare (SW).

8.1. Scenario 1: Single-Segment DDoS Attack

- **Objective:** To verify the basic end-to-end detection and response pipeline within a single network zone.
- **Setup:** A simulated attacker floods the Public-Facing Services segment at 10x baseline volume for 60 seconds.
- **Agents Involved:** Attacker Agents, Traffic Monitor Agent (TMA) , Anomaly Classifier Agent (ACA) , and Response Coordinator Agent (RCA).
- **Expected Behavior:** The TMA must detect the volume anomaly within 100 ms. The ACA must classify the event as a confirmed threat with high severity. The RCA must initiate a traffic throttling directive within 500 ms.
- **Success Criteria:**
 - **Defense:**
 $Detection\ Rate > 90\% \wedge MTTR_{Response} < 1,000\ ms \wedge availability > 99\%$.
 - **Attacker:** $U_{ATK} < 0.2$ (The rapid response must effectively neutralize the attacker's disruption impact).

8.2. Scenario 2: Multi-Segment Coordinated Attack

- **Objective:** To validate the system's ability to correlate threats and form effective defensive coalitions.
- **Setup:** A DDoS attack is launched against the Public-Facing Services simultaneously with a Lateral Movement attack in the Internal User Subnet.
- **Agents Involved:** Attacker Agents, All defense agents, specifically the Threat Intelligence Agent (TIA) and multiple RCAs.
- **Expected Behavior:** The TIA must detect the correlation within 1 second and trigger coalition formation. The coalition must coordinate simultaneous responses across both affected segments.
- **Success Criteria:**
 - **Defence:** Successful coalition formation within 1 second and $SW \geq 0.8$.
 - **Attacker:** $evasion_rate < 0.15$ (The TIA correlation must prevent the Lateral Movement agent from remaining undetected).

8.3. Scenario 3: Resource Contention Under Heavy Load

- **Objective:** To verify the auction-based resource allocation mechanism under high concurrent demand.
- **Setup:** Five simultaneous incidents of varying severity are injected across all segments, with only three available quarantine slots.
- **Agents Involved:** Multiple Attacker Agents, Resource Allocator Agent (RAA) and five RCA instances.
- **Expected Behavior:** The RAA must conduct a sealed-bid auction and allocate slots to the three agents handling the highest-severity threats.
- **Success Criteria:**
 - **Defense:** Auction completion within 300 ms, and system resource usage equal or smaller than 40%.

8.4. Scenario 4: Zero-Day / Novel Attack Detection

- **Objective:** To evaluate the effectiveness of anomaly-based detection against unknown attack signatures.
- **Setup:** A Zero-Day Emulator generates traffic patterns that do not match any known signatures in the ACA's model.
- **Agents Involved:** Attack Agents, TMA and ACA.
- **Expected Behavior:** The TMA must flag the traffic based on baseline deviation. The ACA must utilize its clustering models to classify the event as "Suspicious".
- **Success Criteria:** Detection of the novel attack within 500 ms with a False Positive Rate < 10%.

8.5. Scenario 5: Agent Failure & Resilience

- **Objective:** To demonstrate the system's resilience and its ability to maintain functionality despite individual agent loss.
- **Setup:** The ACA responsible for the Server Zone is forcibly terminated mid-simulation, followed by a localized attack.
- **Agents Involved:** Remaining ACA instances and the TIA.
- **Expected Behavior:** Remaining agents must detect the failure and redistribute duties within 2 seconds to ensure the attack is still processed.
- **Success Criteria:** Coverage re-established within 2 seconds, and $MTTR_{Response}$ remains below 1,000 ms.

8.6. Scenario 6: Voting Protocol Validation

- Objective: To verify that high-risk actions are governed by group consensus through the voting protocol.
- Setup: An RCA proposes a segment quarantine for a moderate-severity threat, triggering a vote within a coalition of five agents.
- Agents Involved: RCA and the active coalition members.
- Expected Behavior: The system must accurately record votes and only execute the quarantine if a majority (>50%) approves.
- Success Criteria: The voting cycle must conclude within 300 ms, with the action outcome strictly following the majority result.